ECA10- MICROPROCESSOR AND MICROCONTROLLER

CO1 CONSTRAINT-BASED PROBLEM-SOLVING ASSESSMENT

Assignment – 1:

Constraint-Based Problem Solving with 8-bit Processor Assembly Language Programming

Course Outcome – CO1

Understand microprocessor architecture and identify different instruction set classifications and addressing modes for arithmetic, logic, data transfer, and branch operations.

Nature of Assessment:

Open-Ended Engineering Problem-Solving / Assembly Programming Assignment
Duration: 1 Week
Maximum Marks: 100

Primary Bloom's Levels: K3 – Apply | K4 – Analyze | K5 – Evaluate | K6 – Create/Design Permitted: Writing the 8-bit Assembly Language Program (ALP) is permitted. All calculations, instruction selection, addressing-mode analysis and expected results shall be worked out manually.

THE ENGINEERING PROBLEM

A simple 8-bit processor-based Monitoring and Control Unit is required for an equipment room. The controller receives two 8-bit sensor values, performs arithmetic and logical operations, compares the processed value with a predefined threshold, and generates a status output. The system must be developed using 8-bit Processor Assembly Language under strict instruction and memory constraints. The specifications are:

Parameter	Requirement / Constraint
Processor	Intel 8-bit microprocessor (Basic Processor)
Input 1	8-bit value at memory location 2050H
Input 2	8-bit value at memory location 2051H
Processing requirement	Add the two 8-bit input values and store the result; compare the processed value with a threshold
Logical operation	Apply a bit mask to the processed 8-bit result

Output status	Memory location 3050H: 00H = NORMAL, 01H = WARNING, 02H = CRITICAL
Arithmetic result	Store the 8-bit arithmetic result at memory location 3051H
Threshold	8-bit threshold value stored at 2052H
Mask	Use a predefined 8-bit mask value stored at 2053H
Register constraint	Use the 8-bit registers A, B, C, D, E, H and L; justify the important register allocation
Program-memory constraint	Target maximum of 256 bytes for the executable logic; report the approximate program size
Instruction constraint	Use at least one instruction from each required class: data transfer, arithmetic, logic, and branch/control
Addressing constraint	Demonstrate the applicable processor addressing modes: immediate, register, direct, register indirect and implied
Verification	Show representative register/memory contents before and after execution for at least three test cases

You are not given:

- the final instruction sequence;
- the register allocation;
- the branch structure; or
- the addressing-mode selection.

You must determine these from the requirements and defend your choices.

Your objective is not merely to produce an ALP.

Your objective is to develop a correct, constrained and explainable 8-bit Processor solution.

STAGE 1 – INTERPRET THE SYSTEM REQUIREMENT

15 Marks | K4 – Analyze

Analyse the engineering requirement before writing the program.

Determine and discuss:

- The role of the 8-bit Processor in the proposed system and the major functional blocks involved.
- How the ALU, Control Unit, registers and buses participate in processing the sensor values.
- Why the 8-bit Processor is suitable for processing the given 8-bit data.
- The required input, processing, decision and output flow.
- The constraints that can influence instruction selection, memory use, register use and program size.
- At least two relevant stages in the evolution from early microprocessors to modern processors, and one engineering improvement introduced by that evolution.

Stage 1 Deliverable

Draw a labelled 8-bit Processor functional-block diagram and conclude with a concise System Requirement Statement.

STAGE 2 – CLASSIFY AND SELECT INSTRUCTIONS

15 Marks | K4/K5 – Analyze and Evaluate

Develop an instruction-selection table for the problem. For each operation below, identify suitable 8-bit Processor instructions, classify them, and justify the selected instruction.

Required operation	Instruction class to identify	Expected evidence
Read temperature/humidity	Data transfer	Instruction + addressing mode + reason
Compare temperature with limits	Arithmetic / compare	Instruction + flags affected
Increment event counter	Arithmetic	Instruction + 8-bit data handling
Generate/clear control bits	Logic	ANA/ORA/XRA/CMA or equivalent
Select NORMAL/WARNING/CRITICAL	Branch / program control	Conditional branch and flag dependency

Copy values to display buffer	Data transfer	Suitable memory addressing
Optional loop over display buffer	Branch / program control	Loop/relative control reasoning

You must also:

- Classify the selected instructions as data-transfer, arithmetic, logical, branch/program-control or other control instructions.
- Explain why at least two alternative 8-bit Processor instructions were rejected under the given constraints.

STAGE 3 – DESIGN THE OPERATIONAL LOGIC

15 Marks | K5 – Evaluate

Construct the decision logic for the controller before writing the complete ALP.

You must:

- Prepare a flowchart or structured decision tree for the NORMAL, WARNING and CRITICAL conditions.
- Define the exact conditions under which the FAN, ALARM and HUMIDIFIER bits are set or cleared.
- Explain the flags used by the comparison and branch instructions.
- Show how arithmetic, logic and branch instructions work together to transform the input data into the required outputs.
- Estimate the minimum number of major processing steps and identify any redundant operation that can be eliminated.

Design Decision 1

State the branch strategy you recommend and justify it in terms of correctness, readability and instruction efficiency.

STAGE 4 – ADDRESSING MODE SELECTION UNDER CONSTRAINTS

15 Marks | K5 – Evaluate

The same task can often be implemented using different addressing modes. Analyse the following modes in the context of this problem:

Addressing mode	Required task	Your analysis
Immediate	Load threshold / mask values	Instruction, operand form, suitability
Register	Intermediate calculations	Instruction and register-allocation reason
Direct	Access 2050H / 2051H / 2052H / 2053H / 3050H / 3051H	Instruction, memory access and suitability
Register Indirect	Access memory through HL	Instruction and benefit
Implied	Accumulator/flag-related operation	Instruction and suitability

Design Decision 2

Select the most appropriate addressing mode for each major data-access operation. Discuss at least two cases where a different addressing mode would work but would be less suitable.

STAGE 5 – DEVELOP THE 8-BIT ASSEMBLY LANGUAGE PROGRAM

25 Marks | K6 – Create/Design

Develop and document the complete 8-bit Assembly Language Program (ALP) for the specified 8-bit controller.

You must:

- Provide an algorithm/pseudocode before the program.
- Provide the complete 8-bit Processor assembly program with meaningful comments.
- Use the required instruction classes and the applicable 8-bit Processor addressing modes.
- Maintain the 8-bit arithmetic and logical processing correctly.
- Store the final status and processed result at the specified memory locations.
- Store the input values and final status at the specified memory locations.
- Clearly identify the role of each 8-bit Processor register used.
- Mark the fetch–decode–execute sequence for at least five representative instructions and explain what occurs at each stage.
- Identify the major instructions contributing to the ALP and explain their selection.

Use of Software Tools

Writing the ALP is permitted; however, no simulator, assembler, emulator, Proteus or other software tool may be used.

The report must contain:

- the reasoning;
- the algorithm;
- the instruction classification;
- the addressing-mode identification;
- the manual calculations and expected results.

STAGE 6 – TEST, VERIFY AND CRITICALLY EVALUATE

15 Marks | K4/K5 – Analyze and Evaluate

Manually trace the ALP for at least the following test cases and verify the outputs. No simulation or software execution is permitted.

Test case	Temperature	Humidity	Expected condition to justify	Record
1	05H	03H	10H	NORMAL
2	08H	09H	10H	WARNING
3	20H	15H	10H	CRITICAL

For each case, show the input memory values, accumulator/register values, relevant flags, arithmetic result, logical result, status byte and output memory contents.

Prepare a final verification table:

Requirement	Specified	Achieved	Satisfied?	Evidence
Correct sensor data acquisition	2050H, 2051H			
Correct condition classification	Defined thresholds			
Correct control bits	3051H			

8-bit arithmetic result	Processed result			
Mask update	3060H onward			
Instruction-class coverage	Data transfer + arithmetic + logic + branch			
Addressing-mode coverage	Applicable 8-bit Processor modes			
ALP correctness and conciseness	Correct 8-bit Processor ALP			

FINAL ENGINEERING DECISION

WOULD YOU USE THESE ROUTINES IN THE SYSTEM?

State clearly:

YES / NO

Defend your decision using evidence obtained from the complete design.

Your conclusion must consider:

- correctness;
- instruction efficiency;
- memory usage;
- register usage;
- addressing-mode suitability;
- verification evidence.

If your answer is NO

Propose a redesign.

Identify exactly what you would change, such as:

- the instructions or addressing modes used;
- the counts, addresses or register assignments;
- the number of registers or procedures;

- the clock frequency;
- the use of additional hardware (timer, counter, DMA controller or interrupt controller).

Explain why you would make the change and predict its effect on the accuracy, speed and reliability of the design and on the rest of the program.

SUBMISSION REQUIREMENTS

Submit one Engineering Design Report.

Your report should demonstrate the complete process:

REQUIREMENT → ARCHITECTURE → INSTRUCTION CLASSIFICATION → OPERATIONAL LOGIC → ADDRESSING-MODE SELECTION → ALGORITHM → ALP DESIGN → EXECUTION-CYCLE ANALYSIS → TESTING → ENGINEERING DECISION

The submission must contain:

- system requirement analysis and 8-bit Processor block diagram;
- instruction classification and selection table;
- flowchart / decision logic;
- addressing-mode analysis and justification;
- algorithm and complete 8-bit Processor ALP;
- register-allocation table;
- fetch–decode–execute analysis for representative instructions;
- manual trace/test evidence and output tables;
- final verification and engineering recommendation.

Listings and screenshots without the corresponding calculations will not receive full marks.

ASSESSMENT RUBRIC

Assessment Criterion	Marks	Primary BL
Interpretation of architecture, system requirements and constraints	15	K4
Instruction-set classification and instruction selection	15	K4/K5
Operational logic and use of arithmetic, logic and branch operations	15	K5

Addressing-mode analysis and selection	15	K5
Development of constrained 8-bit Processor ALP and execution-cycle reasoning	25	K6
Testing, verification and critical engineering evaluation	15	K4/K5
TOTAL	100	

PERFORMANCE EXPECTATIONS

Excellent

The student derives the values systematically, writes correct procedures, calculates the exact timing and addresses, evaluates the alternatives with figures, and verifies the routines against the predictions.

Good

The student follows the correct method and obtains working procedures, with minor errors or limited justification in some areas.

Satisfactory

The student obtains workable procedures but relies mainly on the textbook steps, with limited calculation, comparison or critical evaluation.

Needs Improvement

The student mainly reproduces listings or simulator output without showing the calculations behind the values.